ACTIVITY 7

Aim: To study the modification of root.

Materials Required: Sample / Charts of radish, carrot, beet root, sweet potato, stilt roots and pneumatophores.

Procedure: Carefully observe the shape of each specimen.

Observation: Draw the diagram and observe the morphological differences between the samples.

b. Mechanical Support

Look at a banyan tree. It seems to have many trunk, supporting it. However many of them are actually roots. As the banyan tree is large and huge, it needs support so that it does not tilt and fall down. Many plants require such additional support. Such plants develop roots on their aerial parts to provide mechanical support. These roots grow downward and act as supportive organs. There are three types of modified roots for support.

Prop roots

Roots are modified to provide mechanical support as seen in banyan tree. These roots grow vertically from horizontal branches of a tree.



Banyan

Stilt roots

In sugar cane and maize, adventitious roots arise from the nodes in cluster at the base of the stem. These roots are called stilt roots which give additional support.



Sugar cane

Climbing roots

In betel and black pepper, nodes or internodes bear roots which help in climbing.



Betel



A root growing from a location other than the underground, such as from a stem or leaf is called as adventitious root

VII Std Science Term-1 EM Unit 5.indd 66 23-01-2025 12.49.11 PM